

# Discretization of the Delta Function in the Inhomogeneous Scalar-Advection Equation: Application to Smoke Transport

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# DECLARATION

This work was carried out at AIMS Rwanda in partial fulfilment of the requirements for a Master of Science Degree.

I hereby declare that except where due acknowledgement is made, this work has never been presented wholly or in part for the award of a degree at AIMS Rwanda or any other University.



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# Abstract

Smoke transport can be modeled by a partial differential equation with appropriate boundary and the initial conditions. In this study, the inhomogeneous one-dimensional scalar advection was used with forcing represented by an impulse function. In this case, it is possible to write the solution analytically by using Green's functions and Duhamel's principle. We compare the exact solution to the numerical solution with an emphasis on the direct approximation of the delta function. The resulting equation is solved by Trapezoidal rule on an uniform grid. The accuracy of this method is analyzed for various discretizations and insight gained into the kinds and the importance of choosing adequate discrete delta functions, so that good the accuracy can be achieved.

**Keywords.** Discrete delta function, Scalar-advection equation, Dirac delta function, Analytical solution, Numerical analysis, Smoke, Duhamel's principle, Green's functions.

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# 1. Introduction

Since the industrial revolution, and the demographic explosion, the rate of atmospheric pollution has increased tremendously and this has a significant impact on climate change. Human, animal, and plant health require healthy. In particular, smoke is one of the causes of air pollution due to its different components such as carbon monoxide (CO), oxides of sulfur, nitrogen (SO<sub>2</sub> and NO<sub>x</sub>), carbon dioxide (CO<sub>2</sub>), and all organic pollutants (Sumbane-Prinsloo et al., 2020). It can take any color depending on the initial source and might be easily distinguished by its smell,  $PM_{2.5}$  (Fine particulate matter with diameters 2.5 micrometers and smaller) is a principal pollutant in smoke, and because of its diameters (2.5 micrometers and smaller), it can enter the lungs and bloodstream. Indeed, there is increasing concern about the effects of smoke generated by fires (tires, wildfires, forest fires waste, or wood-burning), (Butler and Mulholland, 2004). In addition, smoke particles can be emitted directly from Biomass burning which affects remarkably air quality, weather, and climate variability. Then, according to Kjellsson (2013), one of the key features of atmospheric circulation is to figure out the way air masses are transported between two different regions. Therefore, the most advanced atmosphere, ocean, and climate models are rely or air quality estimates. Estimating the emission of the particular matter of smoke  $PM_{2.5}$  in the near-surface can be done with one dimension (1D) transport model on an Eulerian grid lead to build the model called transport equation on an Eulerian grid. This transport equation is a simplified version of the Navier Stokes equation found in CDF (Computational Fluid Dynamics) but, estimates rely on observations. Variational data is one of the adequate methods and computationally feasible to combine the PDE (Partial Differential Equation) with data. Variational data assimilation can result in a system of Euler-Lagrange equations. In this essay, the resulting equation is solved analytically and numerically by discretizing a delta function. The Dirac delta function has many interesting properties more and more common (Priyantha et al., 2003). For **Physicist**, a delta function represents a singular force with support only along the boundary. The strength of this force depends on the nature and position of the boundary while for **Mathematicians**, the Dirac delta function replaces the Kronecker delta and takes into consideration the kind of spaces (for instance Euclidean, Sobolev, Hilbert space). Therefore, to reach the expected order of accuracy, the numerical implementation of these delta functions is welcoming and highly recommended by many researchers like C. Peskin through his original work (Peskin, 2002).

Many numerical methods have been applied including the Discontinuous Galerkin Finite Element Method (FEM), Finite Element Methods, finite volume methods (FVM), Finite difference methods (FDM) etc. The selected method for this project is used to replace a partial derivative with a suitable algebraic difference quotient. We consider the one dimensional smoke transport model:

$$\left\{ \begin{array}{l} \underbrace{q_t(x, t)}_{\text{Concentrations}} + \underbrace{\nabla \cdot u(x, t)q(x, t)}_{\text{Velocity advection}} = \underbrace{Q(x, t)}_{\text{Source term}} \quad , \\ q^0(x, 0) = \underbrace{q^0(x)}_{\text{Concentrations}} \end{array} \right. \quad (1.0.1)$$

With periodic boundary conditions.



Variational data assimilation with represents requires numerical solutions of equations (1.3.8)-(1.3.10) in Bennett (1992) where he describe  $\alpha_m$  as a new adjoint variable at the data point  $(x_m, t_m)$  obtained once we have decouple the problem.

Then, the main objective of this work, is to choose the appropriate approximation of the Dirac delta function for solving (1.3.8) in Bennett (2005) For that, we fix some specific objectives such as:

- Decouple Euler Lagrange equations using the representer method
- Explore the use of discrete representations of delta functions
- Determine the analytical solution using Green's function and Duhamel principle
- Estimate the accuracy of the approximations function by using the numerical integration (Trapezoidal rule)
- solve the advection equation using the delta function approximation.

Pope in 1976 developed the probability approach to the modeling of turbulent reacting flows in the space. Base on the nonlinear relations between instantaneous quantities, the transport equations for the single distribution are created, and the ability of the modeled equation to account for the influence of turbulent was demonstrated. For smoke transport, Chow and Yin (2004) proposed a new model for simulating smoke movement induced by fire. Indeed, the Lagrangian approach is used for studying the motion, airflow is simulated by computational fluid dynamics (CFD) using turbulence models. In addition, (Larkin et al., 2010) modeled a particular kind of smoke namely "BlueSky", and presented the overall model's framework showing that the predicted result is affected by the specific choice of modeling pathway.

Tang in 2010 realized that, in a large-scale complicated environment where real-time simulation is hardly achieved, and in virtual simulation, the smoke simulation plays an important role. The MacCormack method is proposed to solve the advection term of Navier-Stokes equation. Rasmussen et al. (2003) presented another method for smoke simulation in case of large scale phenomena. They simulate the motion of particles in a fluid dynamics generated velocity field, and use some physical simulation and continuous mathematics to obtain high resolution. Finally, a numerical study was presented when the phenomenon of advection and diffusion was simulated by varying the parameters (winds and other resources of pollution).

Vira in 2017 presented some applications of the variational assimilation methods in air quality forecasting. A numerical scheme was evaluated to solve the advection equation and then compare to other state-of-art schemes in atmospheric models. She used 4D-Var that requires a forward and adjoint model, and the final result has proved that 4D-Var was more accurate than 3D-Var.

Guerrero in 2013 presented at time dependent analytical solution of the advection-dispersion solute transport equation using Duhamel theorem. The author also developed the generalized equations formulated from the generalized Dirac delta function, evaluated them computationally, and compare them with those in the literature. In addition, Zlatev et al. (2011) also studied the advection equations by applying two numerical method techniques using the Crank-Nicolson

scheme and Richardson extrapolation. They reveal that the accuracy of the numerical solution can increase by applying Richardson extrapolation. The authors conclude that accuracy can be improved considerably if the Crank-Nicolson scheme is combined with the Richardson Extrapolation.

For the special choices of discrete delta function, there exists a number of analysis such as the narrow delta approximations studied by [Beyer and LeVeque \(1992\)](#); [LeVeque and Calhoun \(2001\)](#) who applied it to heat equation in one dimension. To discretize the equation, they used second order finite difference method, and performed a detailed error using the discrete moment conditions. They also suggested that, to carry this type of analysis, various numerical approximations and partial differential equations methods are required. They conclude that different numerical approximation of the differential operator may require automatically  $\delta$ -function regularization.

Najafi and Hajinezhad in 2008 proposed the use of some finite-difference methods to solve the one-advection-dispersion with reaction effect (forward time centered space (FTCS), the backward time, centered space (BTCS), and Crank-Nicolson methods). The authors showed through the plot of the figures that the Crank-Nicolson method works more accurately than the other methods. [Recktenwald \(2004\)](#) provided a numerical solution to the one-dimensional heat equation using two principals methods: The explicit method (FTCS), and the implicit method (BTCS Crank-Nicolson schemes). Results illustrate that, the truncation errors depends on a mesh spacing and time step. This was implemented in MATLAB, and the authors experiment with the stability limit of the FCTS scheme and the choose this method rather than that in [Simon et al. \(2016\)](#) who found that the Crank – Nicolson scheme gives better approximations than FTCS and BTCS. One of the questions about the analytical resolution of the heat equation in the work [Simon et al. \(2016\)](#) was answered by [Subani et al. \(2020\)](#) who used separation of variables to solve analytically the homogeneous partial differential equation. They were able to determine the general and specific solution of heat equation based on the analytical solution. The authors found that the solution of the heat equation depends on the equation itself and the boundary conditions or initial conditions as Neumann boundary conditions.

The flow and motion fluid occurs mostly in engineering and science Typically, applications include biological fluid dynamics the a [LeVeque and Calhoun \(2001\)](#), passively-advected quantity that does not influence the velocity field (transport of smoke, heat, or energy) [Mandli et al. \(2016\)](#), in geological fields through oil and gas extraction [Durst \(2008\)](#). However, a lot of work has been done more precisely on the use of the numerical methods to solve evaluate the analytical solution of the homogeneous and inhomogeneous one-dimensional scalar advection equation. Understanding numerical estimates of the analytical solutions of the one-dimentional scalar-advection equation with a Dirac delta function as a forcing term is addressed in this work.

The remainder of this work is organized as follows: in the second chapter, we present the representer equations that motivate this study, the overview of the Dirac delta function and the Heaviside function which will be follow by the efficient construction of the discrete delta function for different kinds of function (Step function, Hat function, Heat Kernel), and by the numerical analysis while the third chapter is devoted to the computational results. This essay will be closed by a conclusion and some possible suggestions for the future work.

## 2. Euler–Lagrange equations with periodic boundary conditions

In this chapter, we discuss the problem formulation, the governing equations with the associated boundary conditions. The Euler–Lagrange equations are a two-point boundary value problem in time used to solve optimization problems and in this case to determine the change of motion in terms of the space and time. However, when the data about the state of the mechanical system are provided, the mechanical laws for particles is expressed as ordinary or partial differential equations. With  $M$  observations, we have the datum  $d_m$  at the space-time points  $(x_m, t_m)$  on the domain  $(x, t)$  2.1a.

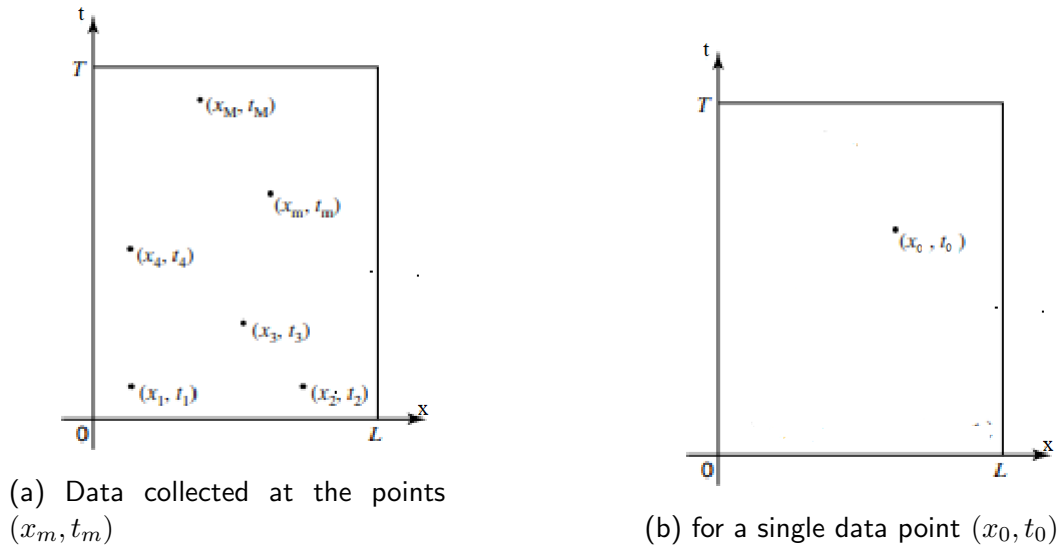


Figure 2.1: model representation (Bennett, 1992)

The Euler–Lagrange equations for local extrema  $\hat{q}$  is given as follow:

$$\left\{ \begin{array}{l} -\frac{\partial \lambda}{\partial t} - c \frac{\partial \lambda}{\partial x} = -w \sum_{m=1}^M \{\hat{q}_m - d_m\} \delta(x - x_m) \delta(t - t_m) \\ \lambda(x, T) = 0 \\ \lambda(L, t) = 0 \end{array} \right. \quad \left\{ \begin{array}{l} \frac{\partial \hat{q}}{\partial t} + c \frac{\partial \hat{q}}{\partial x} = F + \hat{f}(x, t) \\ \hat{q}(x, 0) = I(x) + \hat{i}(x) \\ \hat{q}(0, t) = B(t) + \hat{b}(t) \end{array} \right. \quad (2.0.1)$$

Where :  $d_m$  is the data collected within a spatial domain  $0 \leq x \leq L$  and a temporal domain  $0 \leq t \leq T$ .  $\hat{f}$ ,  $\hat{i}$ ,  $\hat{b}$  the errors terms.  $W_f$ ,  $W_i$ ,  $W_b$  and  $W$  are positive weights choose randomly. Finally,  $F$ ,  $I$ ,  $B$  are the errors in the prior estimates.

the best estimates for  $f, i$  and  $b$  are  $\hat{f}(x, t) = W_f^{-1} \lambda$ ,  $\hat{i}(x) = W_i^{-1} \lambda(x, 0)$ ,  $\hat{b}(t) = c W_b^{-1} \lambda(0, t)$

Loking at equation (2.0.1) we can see that to integrate the first system, it is necessary to evaluate

$\hat{q}(x_m, t_m)$ . Therefore, the Euler–Lagrange equations do not constitute a two-point boundary value problem in the time interval  $0 \leq t \leq T$ .

## 2.1 Representers functions

To assimilate the  $PM_{2.5}$  concentration data directly into the transport equation (1.0.1), we consider weak constraint four dimensional variational data assimilation (4DVAR) with representers. Most variational data assimilation methods that use the strong constraint approach, while 4DVAR is one of them which assumes the existence of the errors terms due to parametrizations and allows data to alter the dynamic in (1.0.1). Weak constraint 4DVAR has a more accurate forecasts and is made computationally feasible using the representer method. for 4DVAR added to equation (1.0.1), results in boundary value problem on  $[0, L] \times [0, T]$  for the adjoint variable  $\lambda(x, t)$  and optimal state estimate  $q$ . Once introduce the representer functions to solve equation (2.0.1), we assume that there are  $M$  observations, denoted by  $r_m(x, t)$ , with  $1 \leq m \leq M$ , each has an “adjoint”  $\alpha_m(x, t)$  satisfying the following equations proved by (Bennett, 1992):

$$\underbrace{\begin{cases} -\frac{\partial \alpha_m}{\partial t} - c \frac{\partial \alpha_m}{\partial x} = \sum_{m=1}^M \overbrace{(q(x, t) - d_m)}^{\text{error}} \delta(x - x_m) \delta(t - t_m) \\ \alpha_m(L, t) = 0 \\ \alpha_m(x, T) = 0 \end{cases}}_{(1)} \quad \underbrace{\begin{cases} \frac{\partial q}{\partial t} + c \frac{\partial q}{\partial x} = \alpha_m(x, t) \\ q(x, 0) = 0 \\ q(0, t) = 0 \end{cases}}_{(2)} \quad (2.1.1)$$

(1): Backward Problem and (2): Forward Problem

We assume that there is a single data point  $\bar{d}$  located at  $t_m = \bar{t}$  and  $x_m = \bar{x}$ , The resulting Euler-Lagrange equation for the Backward Problem is given as below ;

$$\begin{cases} -\frac{\partial \alpha}{\partial t} - c \frac{\partial \alpha}{\partial x} = (\bar{q} - \bar{d}) \delta(x - \bar{x}) \delta(t - \bar{t}) \\ \lambda(x, T) = \lambda(L, t) = 0. \end{cases} \quad (2.1.2)$$

Where  $\bar{q}$  is a concentration of  $PM_{2.5}$  for a single,  $\bar{d}$  a single data point,  $\delta(x - \bar{x})$  ;  $\delta(t - \bar{t})$  represents the Dirac delta functions. So, the equation (2.1.2) is the adjoint equation with the adjoint variable  $\alpha(x, t)$  obtaining by replace the previous adjoint variable  $\lambda$  by the new one ( $\alpha$ ).

Because of the dependence of the two relation above (2.1.2), we will use the linearity and superposition to decouple these problems. For that, we neglecte the error in the system and we replace the backward problem with one which does not involve the expression  $(\bar{q} - \bar{d})$ . we got the following equation:

$$-\frac{\partial \alpha}{\partial t} - c \frac{\partial \alpha}{\partial x} = \delta(x - \bar{x}) \delta(t - \bar{t}) \quad \text{with} \quad \alpha(x, T) = 0 \quad \text{and} \quad \alpha(L, t) = 0. \quad (2.1.3)$$

We will focus on discrete approximations of the delta function when solving the Foward problem (2.1.1), and Duhamel Principle to solve the equation (2.1.3).

## 2.2 Application to an advection equation: Exact Solution

In this Section, we introduce the one-dimensional scalar advection equation with a source term  $\delta$ -function. Then, consider the following advection equation:

$$-\frac{\partial \alpha}{\partial t} - c \frac{\partial \alpha}{\partial x} = \delta(x - \bar{x})\delta(t - \bar{t}). \quad (2.2.1)$$

Subject to

$$\alpha(x, T) = 0 \quad \text{and} \quad \alpha(L, t) = 0.$$

**2.2.1 Green's function for PDEs.** The most simply expression of the PDE problem defining any Green's function is expressed in terms of the *Dirac delta* function. Therefore, solving the inhomogeneous partial differential equation 2.2.1 (Scalar-advection equation with forcing term) needs to seek out the so-called Green's function  $G(x, t; \xi, \tau)$ . The Green's function called  $G$  is a powerful tool used in mathematics to solve some linear inhomogenous PDEs and ODEs and all others equations where initial values can be viewed as boundary values in space-time. The particularity of the Green's function is the uses of the properties Dirac's function. So, we are concerned about inhomogeneous linear PDEs by considering the scalar advection operator defined as  $\mathcal{L} = \frac{\partial}{\partial t} - k \frac{\partial}{\partial x}$  then, the differential equation for  $\alpha(x, t)$  and  $G(x, t; \xi, \tau)$  for  $0 \leq x, \xi \leq L$ ;  $0 \leq t, \tau \leq T$ .

$$\mathcal{L}\alpha(x, t) = f(x, t), \quad (2.2.2)$$

$$\mathcal{L}G(x, t; \xi, \tau) = \delta(x - \xi)\delta(t - \tau). \quad (2.2.3)$$

To solve 2.2.3, we can use the method of separation of variables or Duhamel's Principle which is more efficient than the other classical methods (Hassan, 2017)

**2.2.2 Duhamel's Principle.** As Green's functions, Duhamel's Principle is also a standard techniques to solve inhomogenous linear PDEs. Generally, for finding fundamental solutions, we establish a Duhamel's principle

**2.2.1 Theorem.** (First Order Duhamel's Principle) Concerning problem (2.2.1), for constant velocity define as follow:

$$-\frac{\partial \alpha}{\partial t} - c \frac{\partial \alpha}{\partial x} = Q(x, t). \quad (2.2.4)$$

The resulting solution is:

$$\alpha(x, t) = \alpha(x - ct, 0) + \int_0^t Q(x - c(t - \tau), \tau) d\tau. \quad (2.2.5)$$

In this solution, the first term is often called the *homogeneous solution* while the second term called the *particular solution*. Which is the particular example of the so called *Duhamel's Principle*.

**2.2.2 Lemma.** Given the fundamental solution  $G_0(x, t, \xi, \tau)$  for the equation  $\alpha_t + c\alpha_x = \delta(x - \xi)\delta(t - \tau)$  there are two real  $(x, t) \in (-\infty, +\infty) \times [0, T]$  such that,  $G_0$  is the resulting solution:  $G_0(x, t, \xi, \tau) = \alpha(x, t)$ .

*Proof.* We verify by directly evaluating  $\int_0^t Q(x - c(t - \tau), \tau) d\tau$  since  $\alpha(x - ct, 0) = 0$ . Replacing the two terms  $x$  and  $t$  by:

$$\begin{cases} x \leftarrow x - c(t - \eta) \\ t \leftarrow \eta. \end{cases} \quad (2.2.6)$$

We have:

$$\begin{aligned} \int_0^t Q(x - c(t - \eta), \eta) d\eta &= \int_0^t Q(x - c(t - \eta), \eta) d\eta, \\ &= \int_0^t \delta(x - \xi - c(t - \eta)) \delta(\eta - \tau) d\eta. \end{aligned}$$

□

Using the Shifted heaviside property, we obtained the fundamental solution  $G_0$  see (Bennett, 1992) for more details.

$$G_0(x, t, \xi, \tau) = \delta(x - \xi - c(t - \tau))H(t - \tau). \quad (2.2.7)$$

Where  $G_0(x, t, \xi, \tau)$  is an exact solution by Green's function of the equation (2.2.1)

## 2.3 Discretization

**2.3.1 Discrete delta function.** We usually solved the scalar advection equation (2.1.3) numerically by using numerical methods. However, to obtain the analytical solution of the partial differential equation 2.1.3. We need to know some properties about the Dirac delta function. We often want to represent the delta function as a continuous function that can be evaluated. In particular, it will be important to be able to evaluate  $\delta(x - x_m)$  for non-grid aligned data  $(x_m, t_m)$  figure 2.1a.

**2.3.2 Definition:.** The Dirac delta function is not a function at all, but a distribution or a measure because it can be defined as the 'limit of a sequence of functions' (2.3.1). However, in physics, the Dirac delta function represent a "point source". It can generally be considered as a special function according to its characteristics such as: zero "0" for width and one "1" for area.

$$\delta(x - x_m) = \lim_{\varepsilon \rightarrow \infty} \delta_\varepsilon(x - x_m) = \begin{cases} 0, & x \neq x_m \\ +\infty, & x = x_m \end{cases} \quad (2.3.1)$$

Indeed,  $\delta(x)$  is defined in terms of its property :  $\int_{-\infty}^{\infty} \delta(x) dx = 1$

**2.3.3 . Properties and applications of the Dirac Delta and Heaviside function:** The written  $\delta(x - x_m)$  can also sometimes be written as  $\delta(x, x_m)$ ;  $\delta_m(x)$  or  $\delta_0(x - x_m)$ . The Dirac delta function is also called the unit impulse function, which satisfies several properties. Mathematically, the Dirac delta function is not really a function, but rather a "distribution" that cannot be evaluated directly. For that, we can describe as the delta function  $\delta(x)$  as a "linear functional" with the following properties :

For all continuous functions  $f$ , the **Keys Properties** are:

$$\int_{-\infty}^{\infty} \delta(x - x_m) f(x) dx = f(x_m) \quad (2.3.2)$$

and

$$\int_a^b \delta(x - x_m) f(x) dx = \begin{cases} f(x_m) & \text{if } x_m \in [a, b], \\ 0 & \text{if } x_m \notin [a, b]; \end{cases} \quad (2.3.3)$$

**2.3.4 Heaviside functions.** The Heaviside function plays an important role in the analysis. It is a discontinuous function whose derivative is a Dirac delta function. This discontinuity usually brings numerical issues when evaluating the derivatives. In addition, we often encounter the so-called error function  $\text{erf}(x)$  when working with this function. The rule of the error function here is to make the transition from the Heaviside function to an algebraic function. Therefore, analyzing the various aspects of the Heaviside function is very important. In Table 2.5.1, we display and summarise the mains properties of the Dirac Delta, their mathematical formula and their possible nomenclature.

Table 2.1: Properties of the Dirac delta function and the related Heaviside step function. (Reimers, 2014)

| Property                        | Formula                                                                                                         | Nomenclature                    |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------|
| Normalization                   | $\int_{-\infty}^{\infty} \delta(\xi) d\xi = 1$                                                                  |                                 |
| Indefinite anti-derivative      | $\int_{-\infty}^x \delta(\xi) d\xi = H(x)$<br>$H(x) = \int_{-\infty}^x \delta(\xi) d\xi$ or $H'(x) = \delta(x)$ | Heaviside step function         |
| Integral Origin shift           | $\int_{-\infty}^{+\infty} \delta(x - \xi) dx = H(x - \xi)$                                                      | Shifted heaviside step function |
| Sifting or translation property | $\int_{-\infty}^{\infty} f(\xi) \delta(\xi - x) d\xi = f(x)$                                                    | Shifted ramp function           |
| Scaling                         | $\delta(ax) = \frac{1}{ a } \delta(x)$                                                                          |                                 |
| Ramp function definite integral | $\int_0^1 H_j^i(x) dX = \frac{1}{2} (1 - x_j) \quad 0 \leq x_j \leq 1$                                          |                                 |

**2.3.5 Applications of the discrete Delta function.** In numerical methods, we often want to represent the delta function as a continuous function that we can evaluate. There are several formulas for discrete delta functions  $\hat{\delta}_\varepsilon(x)$  for  $\varepsilon \ll 1$ , three of them are introduced and discussed below according on their properties. Therefore, an explicit form for the Dirac delta function is mathematically given in terms of a sequence of conventional functions.

### 1. Step function

Is a mathematical function defined by

$$\hat{\delta}_\varepsilon(x) = \begin{cases} \frac{1}{\varepsilon} & -\frac{\varepsilon}{2} \leq x < \frac{\varepsilon}{2} \\ 0 & \text{otherwise} \end{cases} \quad (2.3.4)$$

One simple approximation of the delta function is the step function given by (2.3.4) and can also be expressed in terms of a Heaviside function such as:

$$\hat{\delta}_\varepsilon(x) = \frac{H(x + \varepsilon/2) - H(x - \varepsilon/2)}{\varepsilon}. \quad (2.3.5)$$

### 2. Hat function

Also called triangular function, it is a function whose graph takes the shape of a triangle.

$$\hat{\delta}_\varepsilon(x) = \begin{cases} \frac{2}{\varepsilon}(1 - \frac{|2x|}{\varepsilon}) & -\frac{\varepsilon}{2} \leq x < \frac{\varepsilon}{2} \\ 0 & \text{otherwise} \end{cases} \quad (2.3.6)$$

### 3. Heat kernel

It is the fundamental solution to the heat equation given as:

$$\hat{\delta}_\varepsilon(x) = \frac{1}{\sqrt{4\pi\varepsilon}} e^{-x^2/4\varepsilon} \quad (2.3.7)$$

*The plots of these different functions is given in [chapter 3](#).*

In fact, The three latest discrete functions verify the main property of the Dirac delta function mentioned above in the table (2.5.1)

All the properties above would be verified here.

### 1. Step function

[i] **Normalisation**

$$\int_{-\infty}^{+\infty} \hat{\delta}_\varepsilon(x) dx = \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} \hat{\delta}_\varepsilon(x) dx = \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} \frac{1}{\varepsilon} dx = \frac{1}{\varepsilon} \left[ \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \right] = 1 \quad (2.3.8)$$

Therefore, the delta function normalization in this case holds.



[ii]**Scaling** The scaling property holds for each discrete function also if  $\hat{\delta}_\varepsilon(ax) = \frac{1}{|a|}\hat{\delta}_\varepsilon(x)$ . For that, it is more easier to shows that  $\int_{-\infty}^{\infty} |a|\hat{\delta}_\varepsilon(ax)dx = 1$  To prove that, we consider the piecewise integration.

$$\int_{-\infty}^{\infty} |a|\hat{\delta}_\varepsilon(ax)dx = \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} |a| \frac{1}{\varepsilon} dx = \frac{|a|}{a} \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} \xi d\xi = \frac{|a|}{a} \frac{1}{\xi} [\xi] = \frac{|a|}{a} = \begin{cases} if; a > 0 : \frac{|a|}{a} = 1 \\ if; a < 0 : -\frac{|a|}{a} = 1 \end{cases} \quad (2.3.9)$$

Therefore, the step function scaling property holds for the Set function.

$$\int_{-\infty}^{\infty} |a|\hat{\delta}_\varepsilon(ax)dx = 1 \Rightarrow \hat{\delta}_\varepsilon(ax) = \frac{1}{|a|}\hat{\delta}_\varepsilon(x)$$

## 2. Hat function

[i] **Normalisation** Following the same process as in the case of the Step function, we got:

$$\int_{-\infty}^{+\infty} \hat{\delta}_\varepsilon(x)dx = \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} \hat{\delta}_\varepsilon(x)dx = \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} \frac{2}{\varepsilon} \left(1 - \frac{|2x|}{\varepsilon}\right) dx = \frac{2}{\varepsilon} \left(\frac{\varepsilon}{4}\right) + \frac{2}{\varepsilon} \left(\frac{\varepsilon}{4}\right) = 1$$

[iii] **Scaling**

$$\int_{-\infty}^{\infty} |a|\hat{\delta}_\varepsilon(ax)dx = |a| \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} \frac{2}{\varepsilon} \left(1 - \frac{|2ax|}{\varepsilon}\right) dx = \frac{|a|}{|a|} \int_{-\frac{\varepsilon}{2}}^{\frac{\varepsilon}{2}} \frac{2}{\varepsilon} \left(1 - \frac{|2\xi|}{\varepsilon}\right) d\xi = \frac{|a|}{|a|} = 1 \quad (2.3.10)$$

Then, we can conclude that, in this part, both of the properties hold (Normalization and scaling properties).

## 3. Heat Kernel

[i] **Normalisation**

$$\int_{-\infty}^{\infty} \hat{\delta}_\varepsilon(x)dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{4\pi\varepsilon}} \exp\left[-\frac{x^2}{4\varepsilon}\right] dx = \frac{1}{\sqrt{4\pi\varepsilon}} \int_{-\infty}^{\infty} \exp\left[-\frac{x^2}{4\varepsilon}\right] dx = 1 \quad (2.3.11)$$

[iii] **Scaling** In this case, we Step:  $\xi = ax \Rightarrow x = \frac{\xi}{a}$ , and we obtained two different cases:

$$\int_{-\infty}^{\infty} |a|\hat{\delta}_\varepsilon(ax)dx = \int_{-\infty}^{\infty} |a| \frac{1}{\sqrt{4\pi\varepsilon}} \exp\left(-\frac{(ax)^2}{4\varepsilon}\right) dx = \begin{cases} if; a > 0 : \frac{|a|}{a} \int_{-\infty}^{\infty} \hat{\delta}_\varepsilon(\xi)d\xi = 1 \\ if; a < 0 : -\frac{|a|}{a} \int_{-\infty}^{\infty} \hat{\delta}_\varepsilon(\xi)d\xi = 1 \end{cases} \quad (2.3.12)$$

Because  $\int_{-\infty}^{\infty} |a|\hat{\delta}_\varepsilon(ax)dx = 1 \Rightarrow \hat{\delta}_\varepsilon(ax) = \frac{1}{|a|}\hat{\delta}_\varepsilon(x)$ , then the scaling property holds.

### 2.3.6 . Direct Discretization and the approximate Heaviside

In this essay, the numerical regularization have been divided into two steps in order to simplify the analysis. First, the regularized differential equation present in the left hand side is discretized where as in the right hand size , the singular functions are regularized. Therefore, we can approximate the regularized problem by the variety of standard numerical methods. For that, it is essential to consider the direct discretization. We start by approximating the scalar advection equation by considering the  $\delta$ -function source in one dimension, and then, we will show that it is possible to have a perfect approximation for this kinds of problems. By using the trapezoidal rule, we approximate the equation 2.1.3 where we introduce  $\delta_\varepsilon(\xi)$  as an approximation to the delta function and support  $[-\varepsilon/2, \varepsilon/2]$ . In computational, it is most practical to deal with continuous  $\delta_\varepsilon(\xi)$  functions.

By denoting  $x = -\varepsilon/2 + \varepsilon\eta$  where  $0 \leq \eta \leq 1$ , in  $[-\varepsilon/2, \varepsilon/2]$ , we registered for each functions:

- **Step function**

From the above formulas of Heaviside, the corresponding Heaviside function  $H_\varepsilon(x)$  for the Step function is given by the following equation:

– **Case 1:**  $x < -\varepsilon/2$

This case is generally easy.

$$H_\varepsilon(x) = 0. \quad (2.3.13)$$

– **Case 2:**  $-\varepsilon/2 \leq x \leq \varepsilon/2$

In this case, we first compute the integral, and then, by taking the limit as  $\varepsilon \rightarrow 0$ , we realised that it seems like the integral diverges.

$$\int_{-\infty}^x \hat{\delta}(\xi) d\xi = \int_{-\varepsilon/2}^x \hat{\delta}(\xi) d\xi = \frac{1}{\varepsilon} \left( x + \frac{\varepsilon}{2} \right) = H_\varepsilon(x). \quad (2.3.14)$$

So, to fix that, we took into account the fact that  $x$  is also going to zero as  $\varepsilon \rightarrow 0$ , since

$-\varepsilon/2 \leq x \leq \varepsilon/2$ . We can get the true limiting behavior by making the substitution  $x = -\varepsilon/2 + \varepsilon\eta$ , for  $\eta \in [0, 1]$ . After substitution, we get the equivalent problem

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{\varepsilon} \left( \left( \frac{-\varepsilon}{2} + \varepsilon\eta \right) + \frac{\varepsilon}{2} \right) = \eta. \quad (2.3.15)$$

Therefore, The limiting behavior remains finite and in fact is in  $[0, 1]$ .

– **Case 3:**  $x > \varepsilon/2$

This is not so complicated since the function is in the simple form

$$\int_{-\infty}^x \hat{\delta}(\xi) d\xi = \int_{-\varepsilon/2}^{\varepsilon/2} \frac{1}{\varepsilon} d\xi = 1. \quad (2.3.16)$$

Therefore, from the above, the corresponding Heaviside function  $H_\varepsilon(x)$  of the Step function is

$$H_\varepsilon(x) = \begin{cases} 0 & x < -\varepsilon/2 \\ \frac{x}{\varepsilon} + \frac{1}{2} & -\varepsilon/2 \leq x \leq \varepsilon/2 \\ 1 & x > \varepsilon/2 \end{cases} \quad \text{and} \quad \lim_{\varepsilon \rightarrow 0} H_\varepsilon(x) = \begin{cases} 0 & x < -\varepsilon/2 \\ \eta & -\varepsilon/2 \leq x \leq \varepsilon/2 \\ 1 & x > \varepsilon/2 \end{cases} \quad (2.3.17)$$

### • Hat function

Another simple and commonly used approximation of the delta function is the hat function given by (2.3.6). The corresponding Heaviside function  $H_\varepsilon(x)$  is given following the same technique as above. But in this case, we considered two integrals for Case 2 (respectively case 2a and case 2b) case 1 and case 3 remain the same as above (2.3.13), and (2.3.16).

#### – Case 2a : $-\varepsilon/2 \leq x < 0$

In this case, the integral can be written as:

$$\int_{-\infty}^x \hat{\delta}(\xi) d\xi = \int_{-\frac{\varepsilon}{2}}^x \hat{\delta}(\xi) d\xi = \int_{-\frac{\varepsilon}{2}}^x \frac{2}{\varepsilon} \left(1 + \frac{2\xi}{\varepsilon}\right) d\xi = \frac{1}{2} + 2\frac{x}{\varepsilon} + 2\frac{x^2}{\varepsilon^2}. \quad (2.3.18)$$

Therefore, we followed the same trick used in Case 2 (2.3.14) for the step function to evaluate the limit.

Then,

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^x \hat{\delta}_\varepsilon(\xi) d\xi = \lim_{\varepsilon \rightarrow 0} \left\{ \frac{1}{2} + 2\frac{\xi}{\varepsilon} + 2\frac{\xi^2}{\varepsilon^2} \right\} = 2\eta^2 \quad (2.3.19)$$

#### – Case 2b: $0 \leq x < \varepsilon/2$

$$\int_{-\infty}^x \hat{\delta}(\xi) d\xi = \int_{-\frac{\varepsilon}{2}}^0 \frac{2}{\varepsilon} \left(1 + \frac{2\xi}{\varepsilon}\right) d\xi + \int_0^x \frac{2}{\varepsilon} \left(1 - \frac{2\xi}{\varepsilon}\right) d\xi = \frac{1}{2} + 2\frac{x}{\varepsilon} - 2\frac{x^2}{\varepsilon^2} \quad (2.3.20)$$

We also followed the same trick using in Case 2 (2.3.14) for the step function to evaluate the limit.

Then,

$$\lim_{\varepsilon \rightarrow 0} \int_{-\infty}^x \hat{\delta}_\varepsilon(\xi) d\xi = \lim_{\varepsilon \rightarrow 0} \left\{ \frac{1}{2} + 2\frac{\xi}{\varepsilon} - 2\frac{\xi^2}{\varepsilon^2} \right\} = -1 + 4\eta - 2\eta^2, \quad (2.3.21)$$

$$H_\varepsilon(x) = \begin{cases} 0, & x < -\varepsilon/2 \\ \frac{1}{2} + \frac{2x}{\varepsilon} + \frac{2x^2}{\varepsilon^2}, & -\varepsilon/2 \leq x \leq 0 \\ \frac{1}{2} + \frac{2x}{\varepsilon} - \frac{2x^2}{\varepsilon^2}, & 0 \leq x \leq \varepsilon/2 \\ 1, & x > \varepsilon/2. \end{cases} \quad (2.3.22)$$

Set,  $x = -\varepsilon/2 + \varepsilon\eta$  where  $0 \leq \eta \leq 1$ , we obtained:

$$\lim_{\varepsilon \rightarrow 0} H_\varepsilon(x) = \begin{cases} 0, & x < -\varepsilon/2 \\ 2\eta^2, & -\varepsilon/2 \leq x < 0 \\ -1 + 4\eta - 2\eta^2, & 0 \leq x < \varepsilon/2 \\ 1, & x > \varepsilon/2. \end{cases} \quad (2.3.23)$$

### • Heat Kernel

The last function is the Heat Kernel given by (2.3.7), the corresponding Heaviside function  $H_\varepsilon(x)$  is given as:  $H_\varepsilon(x) = \frac{1}{2} + \frac{1}{2}\text{erf}(x)$  with

$$\begin{cases} \lim_{x \rightarrow -\infty} \frac{1}{2} + \frac{1}{2}\text{erf}(x) = -1 \\ \lim_{x \rightarrow +\infty} \frac{1}{2} + \frac{1}{2}\text{erf}(x) = 1 \end{cases}$$

$\text{erf}(x)$  is a special function called: error function (Chevallard, 2012) Then, for  $\eta = \xi/\sqrt{4\varepsilon}$  we have  $x/\sqrt{4\varepsilon}$  and the final result becomes:

$$H_\varepsilon(x) = \int_{-\infty}^x \hat{\delta}_\varepsilon(\xi) d\xi = \frac{1}{2} + \frac{1}{2}\text{erf}\left(\frac{x}{\sqrt{4\varepsilon}}\right). \quad (2.3.24)$$

Knowing that,  $\text{erf}(x) + \text{erfc}(x) = 1$  holds. then, there could be more than one formulation depending on the use of  $\text{erf}$  (2.3.24) or  $\text{erfc}$  (2.3.25).

$$H_\varepsilon(x) = \int_{-\infty}^x \hat{\delta}_\varepsilon(\xi) d\xi = 1 - \frac{1}{2}\text{erfc}\left(\frac{x}{\sqrt{4\varepsilon}}\right) \quad (2.3.25)$$

We realised that, for each function, this equation  $\lim_{\varepsilon \rightarrow 0} H_\varepsilon(x) = H(x)$  is verified. So the anti-derivative property holds. Also,  $\lim_{x \rightarrow -\infty} H_\varepsilon(x) = 0$  and  $\lim_{x \rightarrow +\infty} H_\varepsilon(x) = 1$ .

## 2.4 Numerical approximations

Solving the fundamental solution above (2.2.7) is now possible and quite easy. As mentioned in section (2.3.5) and (2.3.5), we can obtain for different functions (Step, Hat functions and Heat Kernel) of the appropriate approximation, the graphs of the fundamental solution  $G_0$ .

The new expression of the fundamental solution  $G_0$  for  $-\frac{\varepsilon}{2} \leq x < \frac{\varepsilon}{2}$  becomes

$$G_0(x, t, x_m, t_m) = \delta_\varepsilon(x - x_m - c(t - t_m))H_\varepsilon(t - t_m) \quad (2.4.1)$$

$$\begin{aligned} C_x &= \left\{ x_i, i = 0, 1, \dots, N \mid x_0 = a_1, x_i = x_{i-1} + h, i = 1, 2, \dots, N, h = \frac{b_1 - a_1}{N}, x_N = b_1 \right\}, \\ C_t &= \left\{ t_n, n = 0, 1, \dots, M \mid t_0 = a, t_n = t_{n-1} + k, n = 1, 2, \dots, M, k = \frac{b - a}{M}, t_M = b \right\}. \end{aligned} \quad (2.4.2)$$

Assume that  $G_0(x_i, t_n, x_m, t_m)$  is the exact solution of (2.4.1) at an arbitrary grid point belonging to the set defined by the grid (2.4.2), we obtained therefore different expressions of  $G_0$  for each case:

### 1. Case 1: Step function

For this case there are two possibilities: One part is to write directly the corresponding discrete delta function and another part is to express the step function in terms of a Heaviside function which is what it used here.

$$G_0(x, t, x_m, t_m) = \frac{1}{\varepsilon} [H(x - x_m - c(t - t_m) + \varepsilon/2) - H(x - x_m - c(t - t_m) - \varepsilon/2)] H_\varepsilon(t - t_m). \quad (2.4.3)$$

### 2. Case 2: Hat function

Here there are two possible values depending on the sign of  $x$  in order to remove the absolute value, and simplify the calculus.

$$G_0(x, t, x_m, t_m) = \frac{2}{\varepsilon} \left( 1 - \frac{2|x - x_m - c(t - t_m)|}{\varepsilon} \right) H_\varepsilon(t - t_m). \quad (2.4.4)$$

### 3. Case 3: Heat Kernel

The expression of the fundamental solution for the Heat Kernel have the following expression:

$$G_0(x, t, x_m, t_m) = \left[ \frac{1}{\sqrt{4\pi\varepsilon}} e^{-(x-x_m-c(t-t_m))^2/4\varepsilon} \right] H_\varepsilon(t - t_m). \quad (2.4.5)$$

## 2.5 General quadrature rule

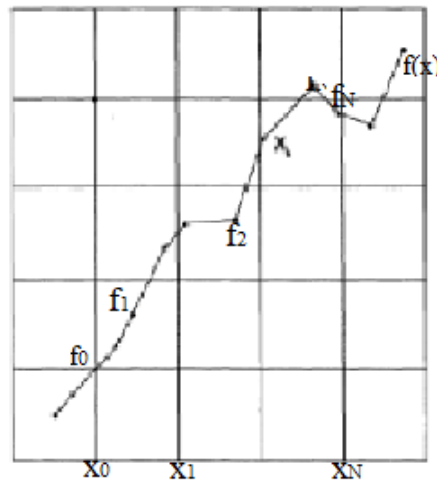


Figure 2.2: Uniform grid (Beyer and LeVeque, 1992)

**2.5.1 Grid for one dimensional model problem.** Another name is the trapezoidal rule approximation. It is a numerical method to be used to approximate the area or the integral under a curve, it is also the exact integral of an interpolate through the given data values. Therefore, approximate the area required to divide it into a number of strips of equal width. To spread the

delta function source to the uniform grid 2.2, we use the discrete delta function  $\hat{\delta}(x)$ . Then, if  $f(x)$  is a function of one variable with the uniform grid spacing, the formula below (2.5.1) is ideal.

1. **Numerical implementation** Let  $\{x_k\}$  be a partition of  $[a, b]$  such that  $a = x_0 < x_1 < \dots < x_{N-1} < x_N = b$  and  $H$  be the length of the  $k$ -th sub-interval ( $H = x_k - x_{k-1} = \delta_x$ ), then we have

$$\int_a^b f(x)dx \approx \sum_{k=1}^N \frac{f(x_{k-1}) + f(x_k)}{2} H = A. \quad (2.5.1)$$

Then, for the domain discretized into  $N$  equally spaced, the considerable simplification may occur:

$$\Delta_{x_k} = H = \Delta_x = \frac{b-a}{N}$$

In this case, the approximation to the integral becomes (?):

$$\begin{aligned} \int_a^b f(x)dx &\approx \frac{\Delta_x}{2} \sum_{k=1}^N (f(x_{k-1}) + f(x_k)) = \frac{\Delta_x}{2} \left( f(x_0) + f(x_N) + 2 \sum_{k=1}^{N-1} f(x_k) \right) \\ &= \Delta_x \left( \sum_{k=1}^{N-1} f(x_k) + \frac{f(x_N) + f(x_0)}{2} \right). \end{aligned}$$

2. **Error analysis** The error of the trapezoidal rule is the difference between the value of the integral and the numerical result. The formula is given by:

$$\text{error} = \int_a^b f(x)dx - \frac{b-a}{N} \left[ \frac{f(a) + f(b)}{2} + \sum_{k=1}^{N-1} f\left(a + k \frac{b-a}{N}\right) \right] \quad (2.5.2)$$

Therefore, there exists one number  $\xi$  between  $a$  and  $b$  such that,  $\text{error} = -\frac{(b-a)^3}{12N^2} f''(\xi)$  and if the number of intervals  $N$  increases to much until reach to  $\infty$ , the error term becomes:

$$\text{error} = -\frac{(b-a)^2}{12N^2} [f'(b) - f'(a)] + O(N^{-3}).$$

**2.5.2 Interpolation properties of discrete delta functions in the scalar advection equation .** If  $f(x)$  is a smooth function of one variable  $x$ , we can also use the discrete delta functions to interpolate  $q(x_j, t)$  grid values to approximate  $q(\alpha, t)$  to the point  $\alpha$ . The discrete form is:

$$f(\alpha) \approx h \sum_j f(x_j) \hat{\delta}_\epsilon(x_j - \alpha) \quad \text{with} \quad f(\alpha) = \int_{-\infty}^{\infty} f(x) \delta(x - \alpha) dx \quad (2.5.3)$$

We conclude that any discrete function defines an interpolation formula. Also, a discrete delta function can be defined by using any translation invariant interpolation. (Beyer and LeVeque, 1992). As  $h \rightarrow 0$  and  $\epsilon \rightarrow 0$  the interpolation error will be at most  $O(h)$ . So, using each of the three discrete delta functions we evaluate the accuracy and the approximation values.

# 3. Numerical Results and Discussions

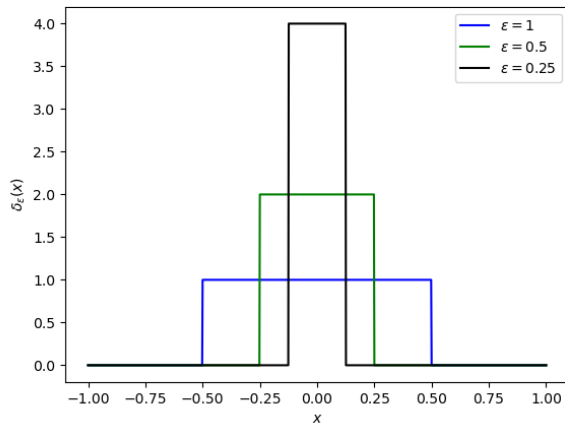
In this chapter, several numerical experiments are performed to validate the previous formulas. For that, all the implementation has been completed in Python 3.

## 3.1 Discrete Delta functions ( $\delta_\varepsilon(x)$ )

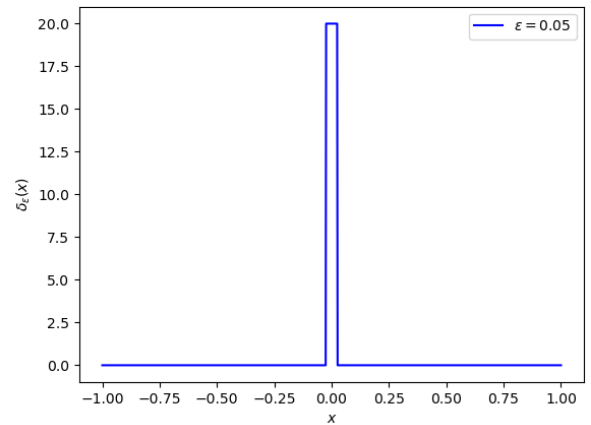
We present the results using the three examples of discrete functions given above ( 2.3.4; 2.3.6; 2.3.7 ) for one dimensional problems. The regularized delta functions is shown by the Graphs below 3.1 ; 3.2; 3.3 respectively for the three discrete delta functions over the interval  $[-1, 1]$  with different values of  $\varepsilon$  on the left hand side and with  $\varepsilon = 0.05$  on the right hand side, and  $N = 1001$  to get enough resolution.

### 1. $\delta_\varepsilon(x)$ from the Step function

The Figure 3.1 shows the Dirac delta function centered at the point  $x = 0$ . it has an open rectangular shape with the value 1 at the bottom. From Figure 3.1a, it is obvious that the smaller the parameter  $\varepsilon$  is, the narrower and higher the pulse. As  $\varepsilon \rightarrow \infty$  the amplitude of pulse becomes very large and its width becomes very small so that for any value of  $\varepsilon$ , the integral of the rectangular pulse will always be 1 which verified perfectly one of the properties of the Dirac delta function (2.3.8). On the other hand, when  $\varepsilon \rightarrow 0$  (3.1b), the pulse height ( $\frac{1}{\varepsilon}$ ) converges to infinity its width becomes very small so that for any value of  $\varepsilon$ . This situation is also verified in (Rapp, 2016; Davenport and Root, 1958) .



(a) Discrete delta function for  $\varepsilon = 1, 0.5$ , and  $0.25$ .

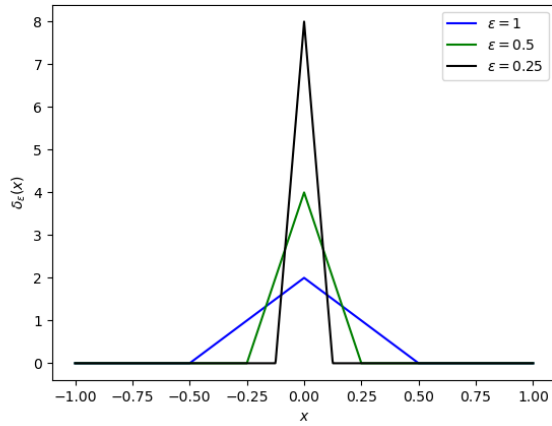


(b) Discrete delta function for  $\varepsilon = 0.05$ .

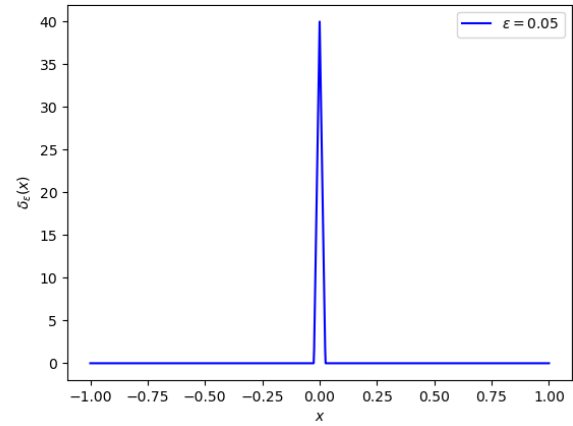
Figure 3.1: Discrete delta for Step function.

### 2. $\delta_\varepsilon(x)$ from the Hat function

The Figure 3.2 demonstrates the approximation of the Dirac delta function. For that Figure, we can see that the values of  $x$  depend on the values of  $\varepsilon$  chosen ( $-\frac{\varepsilon}{2} \leq x \leq \frac{\varepsilon}{2}$ ). Then, the larger  $\varepsilon$  the larger the range of  $x$ . As  $\varepsilon \rightarrow 0$  ( $\varepsilon = 0.05$ ) in Figure 3.2b, the pulse height converges to infinity.  $\delta_\varepsilon(x)$  becomes more and more peaked around the smallest values of  $\varepsilon$  and, become also the limit of increasingly peaked functions. More generally, when  $\varepsilon$  is small, the graph of  $\delta_\varepsilon$  (3.2) is essentially just a spike at  $x = 0$ . In order to obtain a good scheme representing the discrete delta for hat function, it is very important to consider very small values of  $\varepsilon$  as shown in Figure 3.2 which leads to the small values for  $x$  as well.



(a) Delta function for  $\varepsilon = 1, 0.5$ , and  $0.25$ .



(b) Delta function for  $\varepsilon = 0.05$ .

Figure 3.2: Delta function for Hat function.

### 3. $\delta_\varepsilon(x)$ from the Heat Kernel

The Figure 3.3a illustrates the approximation of the delta function. Therefore, at height ( $h = \frac{1}{\varepsilon} \rightarrow \infty$ ), the value of the parameter  $\varepsilon \rightarrow 0$ , and the approximating delta function becomes progressively higher and narrower. In fact, The discrete delta function for the Heat kernel is concentrating at the origin, and following the Gaussian function curve 3.3b developed in (Matignon, 1998; Bennett, 1992) so that, for all elements of the discrete delta function sequence, the above mentioned normalization condition is satisfied. The Figure 3.3, present the approximate Dirac delta function which can get progressively taller and skinnier shape around the smallest values of  $\varepsilon$  in such a way that the area is always 1.

Another important thing to note is the impact of the coefficient  $\varepsilon$  that affects the decay rate. We can see that the decay rate is faster if  $\varepsilon$  is increased (Figure 3.3a)



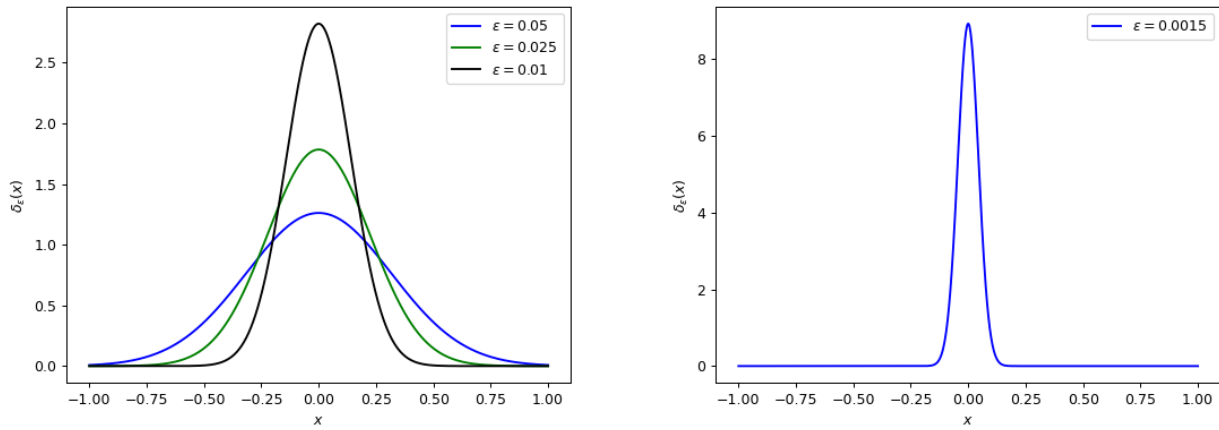
(a) Delta function for  $\varepsilon = 0.05$ ,  $0.025$  and  $0.01$ .(b) Delta function for  $\varepsilon = 0.0015$ 

Figure 3.3: Delta for Heat Kernel.

#### 4. Summary

Looking at all the graph of the three different approximation of the delta function ( $\delta(x)$ ) all depending on  $x$ , we realized that, there is a great similarity in the direction of the different representation which tends to infinity ( $\infty$ ) for very small values of  $\varepsilon$  but in different ways (especially in the shape), such as the step function (3.1b) has at its end a square shape while it is quite different for Hat function (3.2b) which has a pointed shape and for Heat kernel (3.3b) which is just a curved shape and seems like a 1-Dimensional representation of the Gaussian function. At that point, we cannot conclude which is the best one knowing that each one has the particular shape and becomes very interesting when we approximate to the small value of  $\varepsilon$  especially at  $\varepsilon = 0.0015$ .

##### 3.1.1 . Approximation to the Heaviside functions( $H_\varepsilon(x)$ )

In one-dimensional regularized delta functions, we expressed the approximation of the Dirac delta function with a scale-like  $\varepsilon \ll 1$  which is closest to  $[0, 1]$  commonly used in data assimilation. the results obtained for the Heaviside function (2.3.17; 2.3.22; 2.3.24) using the three examples of discrete functions respectively for Step function, Hat function and for Heat kernel are given bellow. The respective graphs should be in the scale  $[-1, 1]$  along the  $x$ - axis with different values of  $\varepsilon$  on the left hand size and with  $\varepsilon = 0.1$  on the right hand size. To have enough resolution, we defined  $N = 1001$ .

##### 1. $H_\varepsilon(x)$ from the Step function

The graph 3.4 below shows the approximation of a unit step function using a discrete delta for four different values of  $\varepsilon$ . We can see from the graph 3.4a that, for the negative values of  $x$ , the function remains steady and take the value '0', while '1' at the positive values of  $x$ . On the contrary, at  $x = 0$ , the set function has exactly the value 0.5 no matter the value of  $\varepsilon$  is. In addition, we realized that, for the smaller the values of  $\varepsilon$ , the more the curve

of the approximate set function fit the one of the Heaviside. Therefore, we considered the value of  $\varepsilon = 0.1$  in the figure 3.4 because we obtained in that condition, a very good approximation of Heaviside.

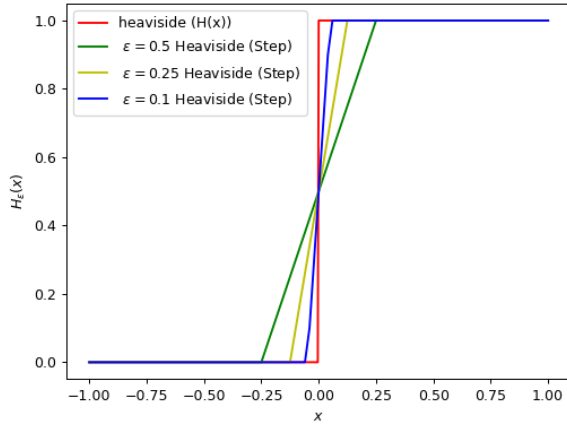
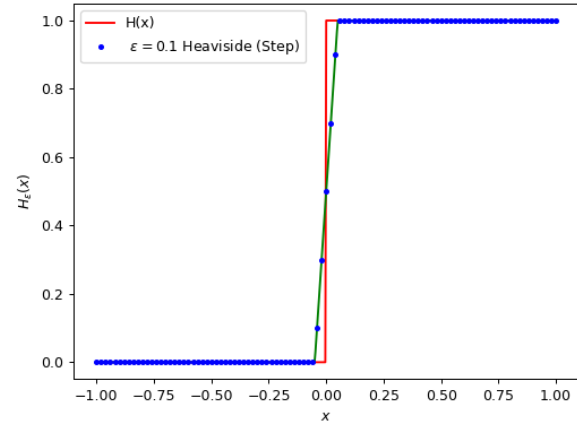
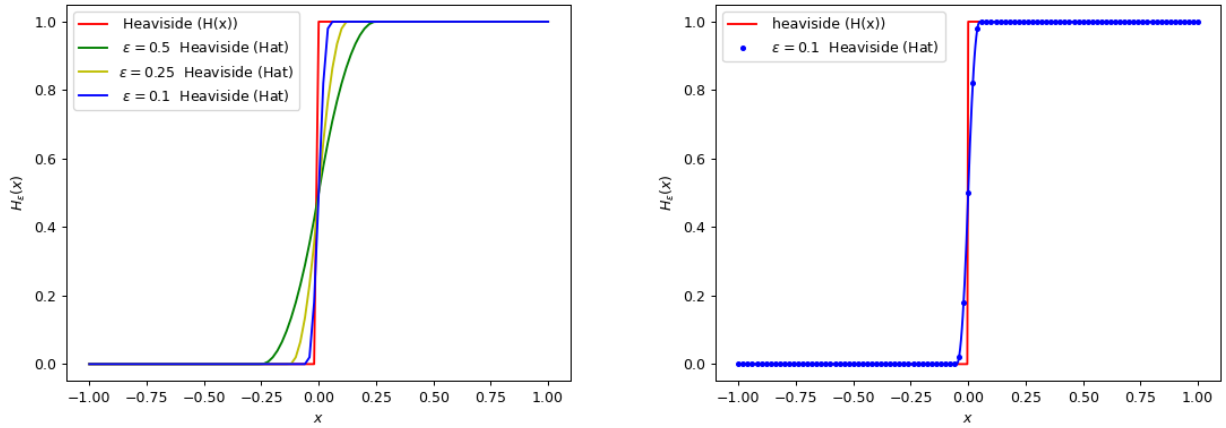
(a) Approximating for different value of  $\varepsilon$ .(b) Approximating for only one value of  $\varepsilon$ .

Figure 3.4: Approximation of a unit step function.

## 2. $H_\varepsilon(x)$ from the Hat function

The Hat functions are differentiable everywhere, and there are piecewise-defined functions as well as the Step function (3.4). Both of them look approximately the same. As for the Step function (3.4b) process, the approximate the Hat function better and better as  $\varepsilon$  gets larger (3.5). If we look at all the derivatives of the Heaviside hat function (2.3.22), we see that the functions are getting narrower and taller, as in the previous case (discrete delta function) see Figure (3.5a). Indeed continuing the process by looking at the ideal  $\varepsilon$ , we found in the Figure (3.5b) an appropriate mode of convergence for  $\varepsilon$  very small ( $\varepsilon = 0.1$ ) the curve of the Heaviside Hat function is close to the one of the non-approximate Heaviside. When a sequence of differentiable functions converges to the unit step, their derivative converges to the delta function. For that, one of the properties of the discrete delta function for the Heaviside function holds (2.5.1). Therefore, it is clear that the derivative of the unit step function can be also defined as the limit of the derivatives.

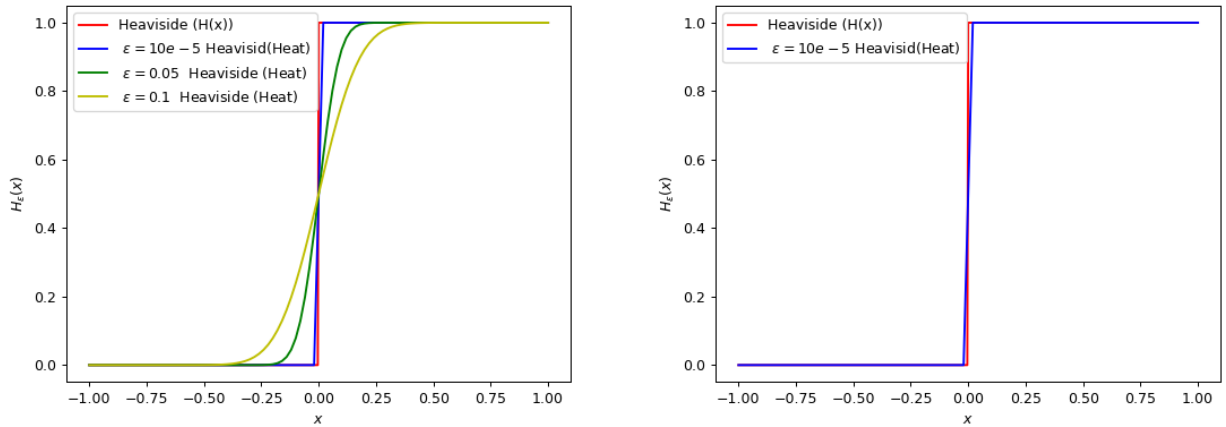


(a) Approximating function for different value of  $\varepsilon$ . (b) Approximating function for only one value of  $\varepsilon$ .

Figure 3.5: Approximation for Hat function.

### 3. $H_\varepsilon(x)$ from the Heat Kernel

The graph below shows the curve of the Heat Kernel approximation of the Heaviside function. Basically, it looks at the same than other functions but to approximate it, we needed to use the smallest value of  $\varepsilon$ . Therefore, the smaller the value of  $\varepsilon$  the better the approximation function looks.



(a) Approximating function for different value of  $\varepsilon$ . (b) Approximating function for only one value of  $\varepsilon$ .

Figure 3.6: Approximation for Heat function.

### 4. Summary

In the light of the above representation of the different approximation of the Heaviside, we realized that the Heaviside function depends on the kinds of the discrete delta function selected which strongly depends to the value of the  $\varepsilon$ . To reduce the accuracy, using

the numerical implementation of the Dirac delta function, we should choose the very small values of  $\varepsilon$ , especially for the case of the Heat Kernel which contains the epsilon term that oblige us to take a very small value to see correctly the approximation function.

## 3.2 Accuracy of the approximation

To evaluate the accuracy of the approximation, we used here each of the three discrete delta function. First, the discrete moment for this model can be following defined as:

$$I(x) = \int_{-\infty}^{\infty} f(\xi) \hat{\delta}_{\varepsilon}(\xi - x) d\xi \approx f(x), \quad (3.2.1)$$

for  $x = 0.45$ , and  $f(x) = \cos(x)$ .

We derive the expression of  $\cos(0.45)$  as follow:

$$\cos(0.45) = \int_{-\infty}^{\infty} \cos(\xi) \hat{\delta}_{\varepsilon}(\xi - 0.45) d\xi \approx \int_{-1}^1 \cos(\xi) \hat{\delta}_{\varepsilon}(\xi - 0.45) d\xi \quad (3.2.2)$$

To approximate the integral (3.2.1), we used the Trapezoidal rule following the different steps mentioned above. For that, we created a mesh on  $[-1, 1]$  with  $N + 1$  equally spaced points. Firstly, we set  $\varepsilon = 0.01$ , and secondly, we set  $\varepsilon = 10^{-5}$  to evaluate the approximation for different functions (Step function, Hat function, and Heat Kernel. Because the value of  $N$  are not fix, then, we Created a table of values for  $N = 16, 32, 64, 128, \dots, 8192$  showing in the tables below:

Table 3.1: Results: Approximate and Accuracy

| $\varepsilon = 0.01$                          |      |                    |            |
|-----------------------------------------------|------|--------------------|------------|
| True value: $\cos(0.45) = 0.9004471023526769$ |      |                    |            |
| $\delta_\varepsilon(x)$                       | N    | Approximate Value  | Error      |
| Step function                                 | 16   | 0.0000000000000000 | 9.0045e-01 |
|                                               | 32   | 0.0000000000000000 | 9.0045e-01 |
|                                               | 64   | 0.0000000000000000 | 9.0045e-01 |
|                                               | 128  | 1.4048178758752163 | 5.0437e-01 |
|                                               | 256  | 1.4074683935922501 | 5.0702e-01 |
|                                               | 512  | 1.0556039797441716 | 1.5516e-01 |
|                                               | 1024 | 0.8796711016819965 | 2.0776e-02 |
|                                               | 2048 | 0.8794638205516543 | 2.0983e-02 |
|                                               | 4096 | 0.8793601275787105 | 2.1087e-02 |
|                                               | 8192 | 0.9013439427328639 | 8.9684e-04 |
| Hat function                                  | 16   | 0.0000000000000000 | 9.0045e-01 |
|                                               | 32   | 0.0000000000000000 | 9.0045e-01 |
|                                               | 64   | 0.0000000000000000 | 9.0045e-01 |
|                                               | 128  | 1.0536134069064185 | 1.5317e-01 |
|                                               | 256  | 0.6149391354100363 | 2.8551e-01 |
|                                               | 512  | 0.9012498264298452 | 8.0272e-04 |
|                                               | 1024 | 0.8793618571874502 | 2.1085e-02 |
|                                               | 2048 | 0.8999564379598194 | 4.9066e-04 |
|                                               | 4096 | 0.8999515671206288 | 4.9554e-04 |
|                                               | 8192 | 0.9005516818618052 | 1.0458e-04 |
| Heat Kernel                                   | 16   | 0.8914362783669935 | 9.0108e-03 |
|                                               | 32   | 0.8914545591283719 | 8.9925e-03 |
|                                               | 64   | 0.8914598776648301 | 8.9872e-03 |
|                                               | 128  | 0.8914612628054888 | 8.9858e-03 |
|                                               | 256  | 0.8914616127075292 | 8.9855e-03 |
|                                               | 512  | 0.8914617004114367 | 8.9854e-03 |
|                                               | 1024 | 0.8914617223517260 | 8.9854e-03 |
|                                               | 2048 | 0.8914617278376928 | 8.9854e-03 |
|                                               | 4096 | 0.8914617292092422 | 8.9854e-03 |
|                                               | 8192 | 0.8914617295521315 | 8.9854e-03 |

Table 3.2: Results: Approximate and Accuracy

| $\varepsilon = 10^{-5}$                      |      |                    |            |
|----------------------------------------------|------|--------------------|------------|
| True value: $\cos 0.45 = 0.9004471023526769$ |      |                    |            |
| $\delta_\varepsilon(x)$                      | N    | Approximate Value  | Error      |
| Step function                                | 16   | 0.0000000000000000 | 9.0045e-01 |
|                                              | 32   | 0.0000000000000000 | 9.0045e-01 |
|                                              | 64   | 0.0000000000000000 | 9.0045e-01 |
|                                              | 128  | 0.0000000000000000 | 9.0045e-01 |
|                                              | 256  | 0.0000000000000000 | 9.0045e-01 |
|                                              | 512  | 0.0000000000000000 | 9.0045e-01 |
|                                              | 1024 | 0.0000000000000000 | 9.0045e-01 |
|                                              | 2048 | 0.0000000000000000 | 9.0045e-01 |
|                                              | 4096 | 0.0000000000000000 | 9.0045e-01 |
|                                              | 8192 | 2.1984090327898489 | 1.2980e+00 |
| Hat function                                 | 16   | 0.0000000000000000 | 9.0045e-01 |
|                                              | 32   | 0.0000000000000000 | 9.0045e-01 |
|                                              | 64   | 0.0000000000000000 | 9.0045e-01 |
|                                              | 128  | 0.0000000000000000 | 1.5317e-01 |
|                                              | 256  | 0.0000000000000000 | 2.8551e-01 |
|                                              | 512  | 0.0000000000000000 | 8.0272e-04 |
|                                              | 1024 | 0.0000000000000000 | 2.1085e-02 |
|                                              | 2048 | 0.0000000000000000 | 4.9066e-04 |
|                                              | 4096 | 0.0000000000000000 | 4.9554e-04 |
|                                              | 8192 | 0.1030504234110479 | 7.9740e-01 |
| Heat Kernel                                  | 16   | 0.0059763888586577 | 8.9447e-01 |
|                                              | 32   | 1.0835949550251456 | 1.8315e-01 |
|                                              | 64   | 0.8751275615364110 | 2.5320e-02 |
|                                              | 128  | 0.9003571088056684 | 8.9994e-05 |
|                                              | 256  | 0.9003570621445269 | 9.0040e-05 |
|                                              | 512  | 0.9003570621445272 | 9.0040e-05 |
|                                              | 1024 | 0.9003570621445272 | 9.0040e-05 |
|                                              | 2048 | 0.9003570621445278 | 9.0040e-05 |
|                                              | 4096 | 0.9003570621445266 | 9.0040e-05 |
|                                              | 8192 | 0.9003570621445270 | 9.0040e-05 |

Interpretation:

First of all, to approximate the integral above (3.2.1) we started by choosing a finite interval of integration  $[-1,1]$  and verified that the value  $x$  is inside. In the case that the interval doesn't contain  $x$ , the integral will always evaluate to '0'. Therefore using the interval  $[-1,1]$  and by taking  $\varepsilon = 0.01$ , we got the table 3.1. Looking at that table, we can see that, there exist different value of the error term depending on the discrete function. As the number of intervals "N" increase,  $h = \frac{b-a}{N}$  is decreased and the error terms as well in all the cases. Which means

that, as  $f \rightarrow 0$ , the approximate become much better estimate of the exact solution. The table 3.1 demonstrates that, for the value of  $\varepsilon = 0.01$  the two functions (Step and Hat functions ) get a value "0" for the three first numbers  $N = 16, 32, 64$  which means that for  $\forall N \leq 16$ , there is no approximation for the integration while for the Heat Kernel, the values of the approximation obtained are closer to the true value  $\cos(0.45)$  with accuracy  $O(h)$ .

On the other hand, the table 3.2 presents the different values of the approximate of the integral using the Trapezoidal rule for each delta function, and for a smallest value of  $\varepsilon$  ( $\varepsilon = 10^{-5}$ ). We can see that  $\forall N \leq 4096$  there is no longer an approximate value of the integral for the Step functions, and for the Hat functions. That means, as  $\varepsilon \rightarrow 0$ , the value of the approximate vanishes in both of the previous functions while in the Heat kernel, the approximate values for the integral ( $I(x)$ ) exist and became more closer and closer to  $\cos(0.45)$  with accuracy  $O(h)$  order 1. This situation is already confirmed by (Peskin, 2002), who said as longer  $h \rightarrow 0$ , the interpolation error will be at most  $O(h)$ .

In the light of the above reason, we can say that the Heat Kernel registered the closest value to the true value than the two other functions (Step and Hat functions). Then, it has a good accuracy and a very good approximation. Therefore, The Heat Kernel is highly recommended for our study.

### 3.3 Numerical Experiments

In this section, we shall solve numerically the Green's function (fundamental solution of the scalar advection equation) for the three different functions and then, the scalar advection equation by considering only the Heat Kernel function as forcing term.

#### 3.3.1 . Green's function: Exact solution

To evaluate the fundamental solution  $G_o$  in the analytical part (2.2.7), we used the quadrature rule. The routine above solves the scalar advection equation for  $N + 1$  equally spaced points of  $x \in [a, b]$ . The solution  $q(x_i, t)$ ,  $i = 0, 1, 2, \dots, N$  is returned in an array, along with an array  $x$  containing the  $x$  values. The figures bellow 3.7, 3.8, 3.9 show the fundamental solution of the one-dimensional advection equation for different functions (Set function, Hat function and Heat Kernel.)

In this work, it is convenient, but not necessary, to divide the time-interval  $[a, b]$  into 0.5 equal sub-intervals. By this convention, in the all three examples of function given in this section, the length of the time-interval becomes  $T = 0.5$ . In each experiment, the run is perform by using  $M = 20$  (time limit number) and  $N = 128$  (spatial limit number)

#### 1. Fundamental solution of the one-dimensional advection equation for Step function

Set:  $M = 20$  ,and  $N = 128$ , we observed the behavior of the graph for different intervals of time one close to  $t_m = 0.1$  and then two more showing the solution growing due to the influence of  $\delta(t - t_m)$  and the last one at  $T = 0.5$  the final time. the solution in time is shown bellow in Figures (3.7 ; 3.8 and 3.9)

In figure 3.7, we can see that, the value of the exact solution increase from 10 to 20 and then continues to evolve without changing the shape and in a constant way along the  $x$ -axis,

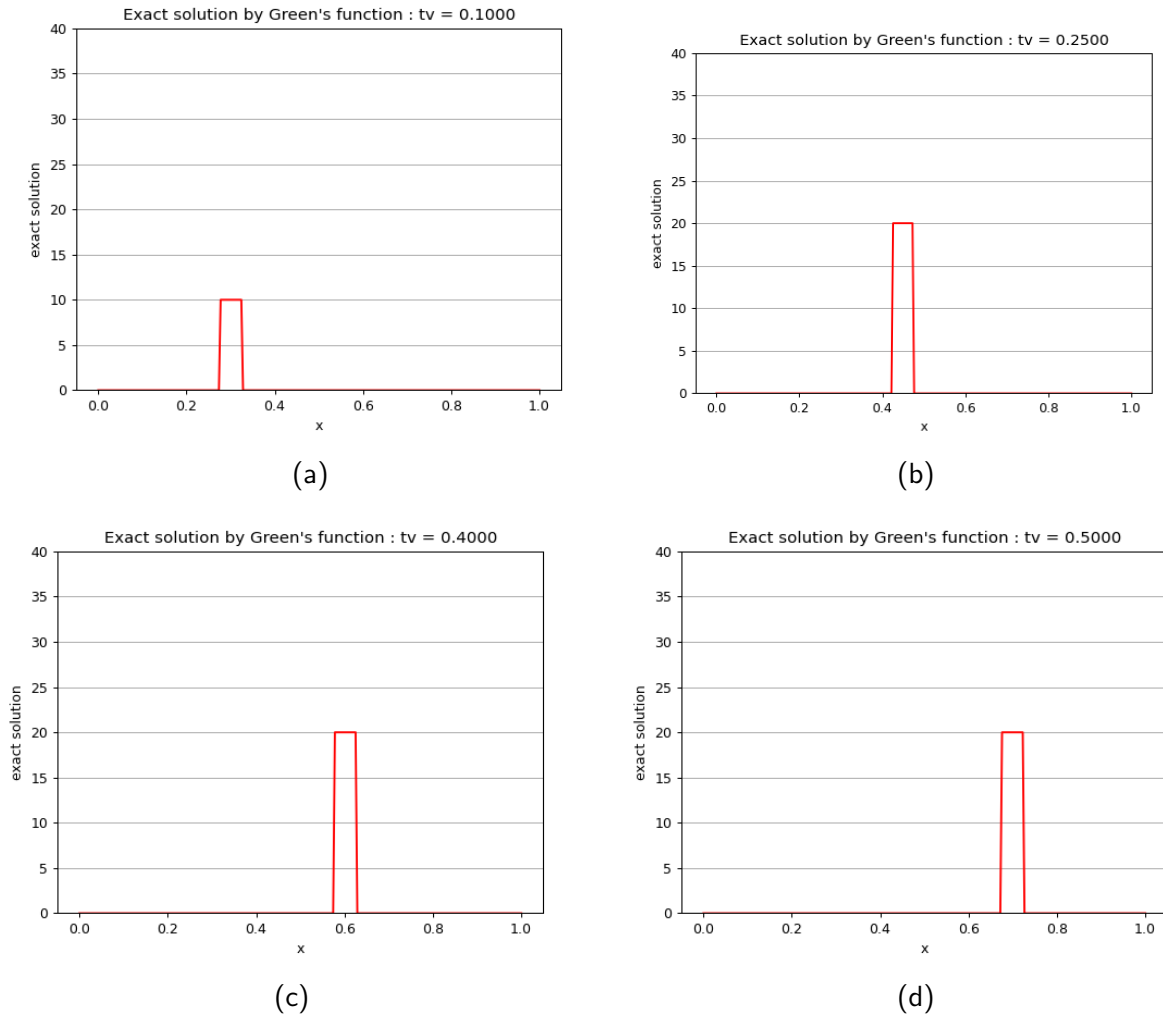


Figure 3.7: fundamental solution of the one-dimensional advection equation defined in equation 3.7. (a) at the beginning. (b) at the middle of the time interval (c) at 1 second left before the end (d) at the end of the time interval.

## 2. Fundamental solution of the one-dimensional advection equation for Hat function

Looking at the previous plot 3.7, the peak location's predicted correctly but the square pulse has been rounded Figure 3.8. We also noticed that the value of the exact solution double after 0.15 second and then, involving in time sinusoidal.

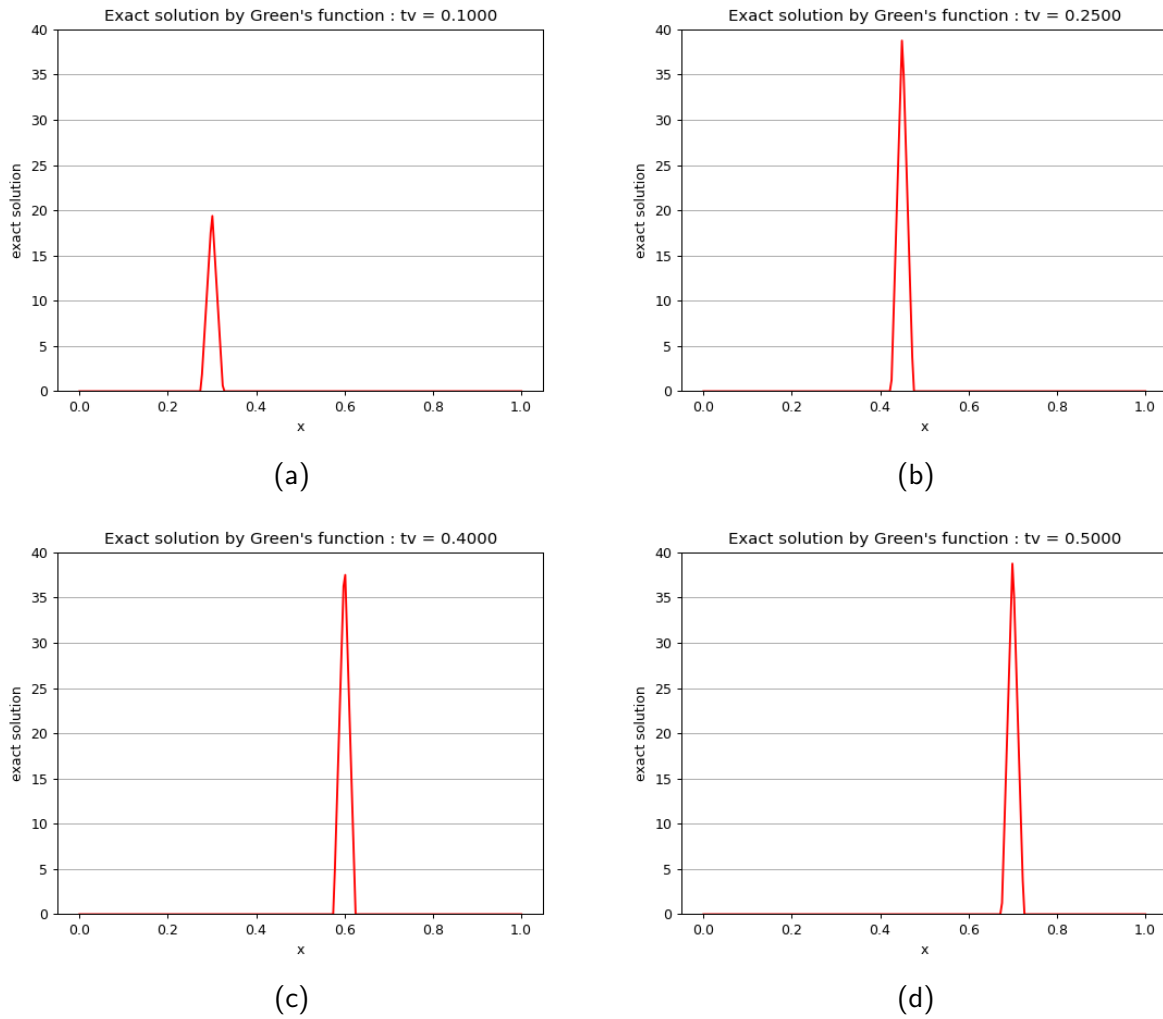


Figure 3.8: Fundamental solution of the one-dimensional advection equation defined in equation 2.4.4. (a) at the beginning. (b) at the middle of the time interval, (c) at 1 second left before the end (d) at the end of the time interval.

### 3. Fundamental solution of the one-dimensional advection equation for Heat Kernel

Near to the initial value, exactly at 0.1 seconds, the exact solution is evaluated around 4.5 and double itself (take a value around 9) just after 0.01 second. However, what we are getting here seems to be more like the square pulse that we started out with. At point 0.6 and 0.8, we can see that it seems gradually sloping hill type of variation moving like a normal distribution. This remark is also highlighting by Georgiev et al. (2011) who used numerical method rather than analytical methods such as *Richardson* Extrapolation and *Crank-Nicolson*.

In Figure 3.9, we can see that the profile moves to the right because ( $c = 1 > 0$ ). Which is the reason that this equation is often called the *the transport equation*. Also, there are two features for the slope such that: positive slope ( means decreasing in time); negative slope ( means increasing in time).



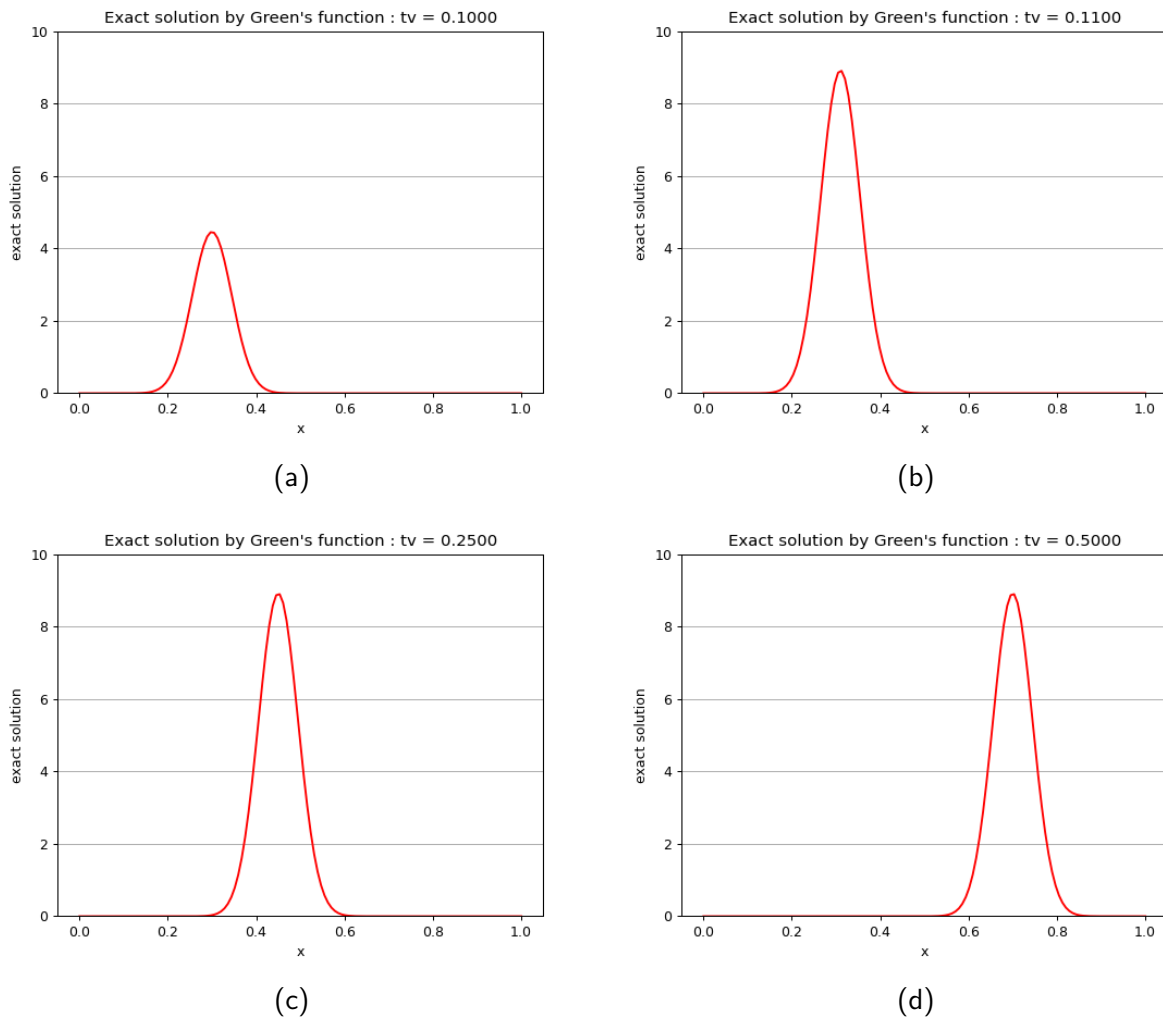


Figure 3.9: Fundamental solution of the one-dimensional advection equation defined in equation 2.4.5 . (a) at the bargaining. (b) at the middle of the time interval, (c) at 1 second left before the end (d) at the end of the time interval.

#### 4. In summary:

In figure 3.7, 3.8, and 3.9 the values of the exact solution at different location at any given time versus the values of the variable  $x$  which are given showing (a) the value close to the initial value, (b) the solution at the middle of the time interval, (c) the solution at at 1 second left before the end (d) the solution at the end of the time interval for the problem defined by 3.7-3.9. So, we have zoomed in on the region between the point 0 and the point 1 for all cases. We notice that, initially, we had a pulse which was centered about point 2.5 and it is moving at 1 meter per 0.1 seconds. Therefore, in 0.1 seconds, it would have moved point 0.2 without changing its shape to the  $x$ -direction. Also, we can see that, in all the cases, the completed pulse is traveling in the positive  $x$ -direction at the correct speed and at the accuracy increasing as the values of  $x$  increased. However, we are expecting better accuracy, with a centered point, because at that point, the time step is

less and worse accuracy.

It is highly desirable to investigate carefully the performance of the forcing term involving the Heat kernel for the test problem studied because it can help to gain in time and it is much stable than others.

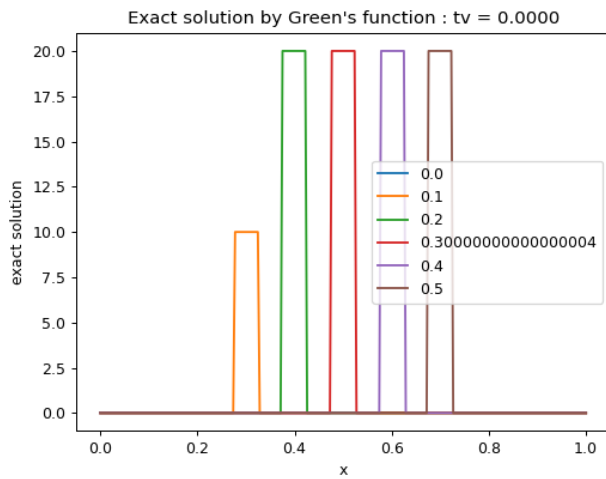
**3.3.2 Solution of the Scalar advection equation for Set function, Hat function and Heat kernel at various times slices.** Plotting together and considering the following parameters:  $M = 5$ , and  $N = 128$  we observe a behavior of the graph for different intervals of time showing by the Figure 3.10 bellow. In this case, we keep the spatial domain to be in the interval  $[0, L]$  with  $L = 1$ , and the same boundary conditions as in (2.1.1).

Looking at the Figure 3.10a, we observe that at  $T = 0.0$  the exact solution take the value "0" and start changing at  $T = 0.1$ . And then, at  $T = 0.2$  up to  $T = 0.5$ , The shape is similar to that of the rectangular function and the value of the solution is remaining the same (20.0). We can directly conclude that as  $T$  increased, the value of the exact solution increase as well up to a certain value and then, remain constant.

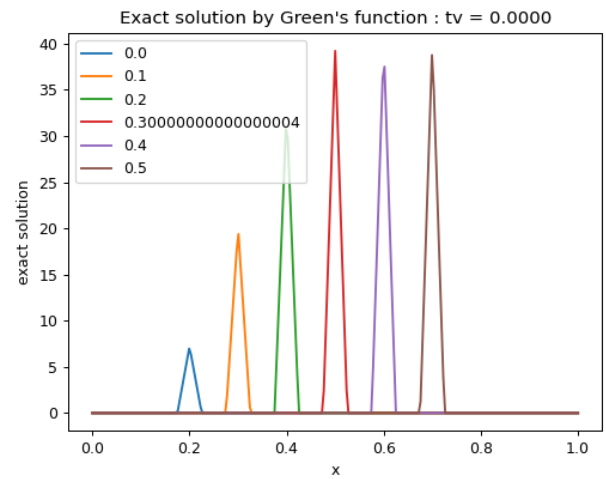
The Figure 3.10b presents the exact solution by Green's function over the different time with a constant speed  $c = 1$  at the interval of time ( $\Delta T = 0.1$ ). It is clearly shown that there is different peak at different time which means that the function evolves as a sinusoidal function.

It is clear that, the graph 3.10c shows the concave up (increasing in time) and concave down (decreasing in time) with the step ( $\Delta T = 0.1$ ) for the time interval  $[0, 0.5]$ . In addition, with the boundary conditions pinned to zero, and the considering the form of the Heat kernel, we might expect  $q \rightarrow 0$  as  $t \rightarrow \infty$ .

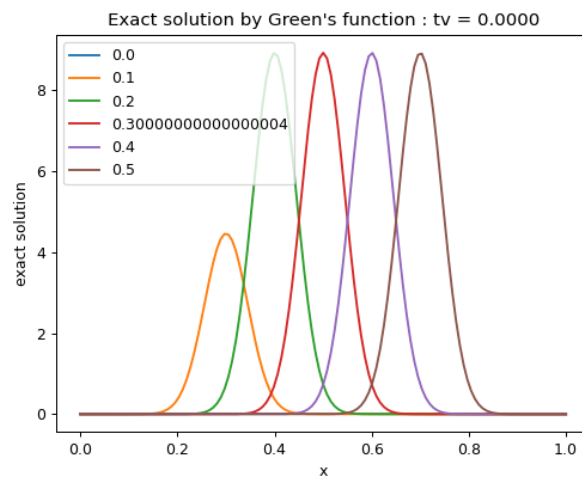
The key observation or the fundamental thing to observe is that, the shape is not changing along a very specific kind of lines (called characteristics lines) for each functions. mathematically,  $q$  is constant on lines  $x - ct = \text{const}$  means that the direction derivative of  $u$  in direction of  $x - ct = x_0$  is zero. This result is confirm in P.11 of the book of (Bennett, 1992).



(a) Step function



(b) Hat function



(c) Heat Kernel

Figure 3.10: Plot of the fundamental solution

## 4. Conclusion and Future Work

Providing an analytical solution to certain equations (Heat equation, advection equation, shallow water equations, ...) present sometime many difficulties in such a way that, a huge research into the improvement another approach to solve those equations has been done and continue to be helpful. In this work, we implemented the representer for solving the adjoint equation. For that, we have developed a discrete delta function that is concentrated on lines in one dimensions time dependant. In fact, because the delta function is in concentrated within one Cartesian grid, we have analysed the accuracy of the regularization of Dirac delta functions which is useful for standard computational grids connected numerically to the differential equations with singular source terms. Three different approximation of the Dirac delta function have been studied such as the Step function, the Hat function ,and the Heat Kernel. The main contributions are in the analysis and development of techniques for this uni-dimensional problems in order to choose the appropriate discrete delta functions and or the practical useful case of regularizations  $\delta_\varepsilon$  with smallest narrow support.

In this essay, we have studied  $\delta_\varepsilon$  functions with a narrow support  $\varepsilon \ll 1$ . We first studied the one dimension, and the discretization error such that a delta approximation is determined by the number of discrete moment conditions (2.1). Based on the order of the discretization, the discrete moment conditions can determine the accuracy with error  $O(h)$  for this study. We have also discussed on different ways to select the best approximation among the three given discrete delta functions.

Firstly, the approach of the numerical integration (quadrature rule) has been used to evaluate the accuracy of the approximation for different functions with  $\varepsilon = 0.01$  ,and  $\varepsilon = 10^{-5}$ . Considered the values obtained, The Heat Kernel present a very good accuracy then, it is the good one. Secondly, solving analytically the scalar advection equation, we obtained the same solution as the book of [Bennett \(1992\)](#) within the same boundary condition, Therefore, to solve the Green's function, we proceed numerically and we found that, the Heat Kernel, presents reasonable solution. Finally, computing the partial differential equation with the source term as an Heat Kernel, reveals the plots that looks like a Gaussian function under a certain conditions, and shows that the Heat Kernel function realizes the expected order of accuracy.

**4.0.1 Future work:.** In the framework of this essay, the results obtained are promising, satisfactory and cover our expectation as well. In order to perform this work in the future, it is highly desirable to investigate carefully the performance of the discrete delta function since the resulting source terms are, indeed, the discrete delta function : Then, we suggest some following ideas:

- Extend the one dimensional  $\delta_\varepsilon$  approximation to several dimensions.
- Studied the case where the velocity  $u$  is no longer the constant but a variable depending on the time ( $u(t)$ ).
- Extend the data point  $(x_m, t_m)$  with  $\leq m \leq M$

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